

Lecture 3 - Exercises

- Exercise 1

Write the exact conversion formula between Unix time and Iso 8601

- Exercise 2

Give (at least) one reason why not to just switch all timing information regarding software, computers, and distributed systems on the ISO 8601 time.

- Exercise 3

Assume that the function `currentTime()` returns the time of your system, which periodically updates via an NTP server and that the function `fun()` takes 50ms to run on your machine. You run the following code:

```
long startTime = System.currentTimeMillis();  
fun();  
long endTime = System.currentTimeMillis();  
long elapsedTime = endTime-startTime;
```

Give bounds on `elapsedTime`.

- Exercise 4

Calculate all the happens-before pairs in the Lamport diagrams on slides 20 and 21.

- Exercise 5

Prove that for two events a and b in a distributed system, Either $a \rightarrow b$, $b \rightarrow a$, or $a \parallel b$.

- Exercise 6

A relation R is a strict partial order if it is irreflexive (not exists event a such that (a,a) is in R) and transitive (for any events a, b, c : if (a,b) in R and (b,c) in R implies (a,c) in R). (These two conditions also imply that it R asymmetric, i.e. that for any events a,b , if (a,b) in R then (b,a) not in R .) Prove that the happens-before relation is a strict partial order.

Hints: You need to prove the two conditions (irreflexive, transitive) separately from the definition of happens-before. To prove some of them you may need to assume that any two nodes are a nonzero distance apart, as well as the physical principle that information cannot travel faster than the speed of light.

- Exercise 7

Assume an NTP client which at time $t_1 = 3:31\text{am}$ sends a request to an NTP server, which arrives at time 3:30 (at the server side). The server then responds with a message "response(3:31, 3:30, 3:31)", which is received by the NTP client at time 3:34. What is the estimated network delay, and what is the estimated clock skew of the client?

Discuss what a clock correction would look like based on the skew you calculated.

- Exercise 8

In a fail-stop model, which of the following properties are safety properties?

1. every process that crashes is eventually detected;
2. no process is detected before it crashes;
3. no two processes decide differently;
4. no two correct (that never crash) processes decide differently;
5. every correct process decides before t time units;
6. if some correct process decides then every correct process decides at some point.